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Episode 54 · November 11, 2026 · 3 papers

Graph-Sparse Distributed Predictive Control for Multi-Robot Motion Planning vianEdge-Consensus ADMM

SKIM

Josep Borras-Borras, Pablo Segovia, Carlos Ocampo-Martinez

Distributed MPC for up to 32 simulated planar robots: they only coordinate with neighbours they are predicted to conflict with, solve one convexified QP per update with edge-consensus ADMM, and fall back to a centralized safety filter when the distributed plan isn't safe enough.

PROBLEM Distributed collision-avoiding MPC usually couples every pair of agents and must re-solve nonconvex constraints. That makes coordination traffic and computation grow badly with team size, and it has kept consensus-based joint planning from scaling.

METHOD Each update, a predicted conflict graph built from warm-start trajectories decides which robot pairs need to interact. Only those pairs' collision constraints are linearized into separating half-spaces, which fixes a single convex QP. Edge-consensus ADMM then splits the QP into parallel per-robot solves plus closed-form per-edge projections. A centralized high-order control barrier function QP checks the result and steps in whenever the distributed candidate fails the direct-execution conditions.

RESULTS The abstract reports C++ simulations with 8, 16 and 32 robots. It claims collision-free execution, much better computational scaling, and task performance comparable to a centralized complete-graph baseline. It gives no runtimes, no iteration counts, no performance gaps, and no figure for how often the centralized filter had to step in.

CAVEAT The safety guarantee comes from a centralized barrier-function supervisor, so the system is not fully distributed. If that filter fires often, the scaling and safety claims mostly reflect the central fallback, and the abstract doesn't say how often it fires. Everything else is idealized too: point-mass double integrators in an open plane, simulation only, and a perfect, synchronous exchange of ADMM iterates with no packet loss, latency or non-participating agents. A reviewer would also ask what fixed ADMM iteration budget was used and how far from consensus the plans are when the solver stops early.

WHY IT MATTERS This is peripheral to your work. It fits under distributed, consensus-based cooperative trajectory planning, and the conflict-graph trick for pruning which pairs must negotiate is a reusable idea for deciding which vehicles need to exchange plans. But it has no lanes, no road structure, no intent or proposal messages, no explicit yielding or deadlock resolution, and no model of degraded communication. Those are the parts your work actually depends on.

arXiv:2611.11755v1 PDF incremental

A Game-Theoretic MPC Planner for Cooperative and Adversarial Drone Coordination

SKIM

Zdeněk Bouček, Miroslav Flídr

A game-theoretic model predictive controller for small drone teams, tested on cooperative goal reaching and pursuit-evasion in PyBullet, with one lab flight that checks only the trajectory-tracking interface.

PROBLEM Game-theoretic planners that model each agent's objective and shared collision constraints are attractive for multi-robot conflict resolution, but their real-time limits in 3D with several agents are poorly characterised. The paper tries to find where such a planner stops being fast enough or numerically reliable.

METHOD Each drone gets its own objective over a 3D prediction horizon, coupled through optional shared separation constraints, and an off-the-shelf cooperative game solver computes the equilibrium trajectories. They compare several game solvers on whether they find feasible, conflict-free trajectories. They use move blocking, which holds control inputs constant over groups of steps, to cut the number of decision variables and the computation time.

RESULTS The abstract gives no numbers at all, so there are no solve times, success rates, agent counts or horizon lengths to quote. It only says that timing measurements identify practical real-time limits, that PyBullet shows both scenarios, and that one lab flight validates trajectory tracking on real drones.

CAVEAT This is a characterisation study of existing solvers, not a new algorithm, and the hardware flight validates tracking rather than the game-theoretic planning itself. Nothing in the abstract suggests the agents exchange messages. The game appears to be solved with full knowledge of every agent's objective and state, so communication loss, delay, partial participation and decentralised execution are never tested. Given seven pages and no numbers in the abstract, the number of drones tested is probably small.

WHY IT MATTERS This is peripheral to your work. It touches multi-agent conflict resolution through shared separation constraints and equilibrium solving, but it assumes complete information instead of negotiation, so it has nothing to say about maneuver messages, proposal and acceptance, or degradation under lossy links. The one reusable piece is the move-blocking timing study, which may help if you need to budget compute for a game-theoretic MPC baseline in vehicle merging.

arXiv:2611.12226v1 PDF incremental

SoftPAT3D: Physics-Augmented Text-to-3D Scene Generation with Deformables

SKIM

Lei Shu et al.

SoftPAT3D makes text-to-3D scenes that contain cloth, stuffed toys and ropes simulation-ready, so those objects settle onto furniture without passing through anything, and it has nothing to do with connected vehicles.

PROBLEM Physics-aware text-to-3D pipelines only handle rigid bodies. A generated deformable object comes back as a mesh that may intersect itself, with no rest shape, no material model and no valid starting state, so it cannot be dropped into a deformable simulator.

METHOD They take the rest shape from how the object would be manufactured, not from the deformed shape the generator produced. A material library supplies starting elastic parameters. The hidden support surface is regenerated as a watertight collision mesh, and every body is released from above its support and settled to static equilibrium under one shared contact energy. Geometric checks and a vision-language model confirm success. When a scene fails, a grid search over placements fixes it, and gradients from a differentiable simulation of the same contact energy handle the cases the grid misses, jointly place several bodies and tune material parameters.

RESULTS Across sixty-five prompts, SoftPAT3D shows no unwanted interpenetration while every baseline shows some. The abstract does not name the baselines and gives no numbers for physical realism, run time or prompt fidelity.

CAVEAT The headline metric is a binary no-interpenetration check, which an equilibrium solver with a barrier-style contact model is built to pass, so it says little about whether the scenes look right or match the prompt. Sixty-five prompts is a small benchmark, prompt adherence relies on a vision-language model judge, and the robotic manipulation claim seems to be a demonstration rather than a measured result.

WHY IT MATTERS This is outside your area. It covers static household scenes with deformable objects and involves no vehicles, communication, negotiation or multi-agent planning. The one loosely related idea, solving a joint placement of several bodies against a shared contact energy, does not carry over to coordinating moving vehicles that exchange messages. You can safely skip it.

arXiv:2611.11087v1 PDF notable